SAAD71000 Object-Oriented Analysis and Design

CA1 Analysis Report

Scenario CA1-24-HEBS: Harbour Equipment and Berth Service

Kerry Harbour Services Cooperative

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23 October 2026

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1. Scenario interpretation

1.1 Scenario evidence

ID	Scenario evidence	Initial observation	Questions (Q) and assumptions (A)	→ Used in
E01	The cooperative coordinates short-term berth use and shared harbour equipment for fishing, leisure and community vessels. Equipment includes mobile fenders, shore-power leads, pumps, lifting accessories and safety barriers.	No problem noted.	None.	Current process
E02	Berth requests, vessel details, equipment reservations, inspections and harbour notices are managed through separate logs and messages.	Information is fragmented. Staff may miss who booked what; records are easy to lose; several people recording may cause confusion (interpretation, not stated directly in the scenario).	Q01: Are the logs paper-based or electronic? Are they kept by one person or by several staff members? Q27: Which staff role records berth requests and their changes, and which role gives out equipment?	P1
E03	Weather, vessel changes and operational restrictions can alter plans quickly.	Plans are unstable.	Q20: What operational restrictions exist, who introduces them, and how are staff and vessel operators informed of them? Q21: Who informs a vessel operator that plans have changed, and how?	P5
E04	A request includes vessel, dimensions, arrival and departure window, purpose, contact and required services. Exact times can change.	No problem noted.	Q02: Which vessel information and evidence is required before a booking is confirmed? Q22: How far in advance must an operator report a changed arrival time, and how is this done? (related to Q04, Q11) Q23: Do vessel operators use the system themselves, or only through harbour staff? Q26: Which staff roles are allowed to see operator contact details? A07: Harbour staff enter berth requests and changes for vessel operators. The operators do not use the system themselves (temporary assumption).	Request content -> requirements, concepts
E05	Berths differ in length, depth, access, power and operational restrictions. A physically empty berth may be unsuitable or reserved within the same time window.	An empty berth is not necessarily a suitable berth.	Q03: How should berth suitability for a vessel be determined?	P2

ID	Scenario evidence	Initial observation	Questions (Q) and assumptions (A)	→ Used in
E06	Some requests remain provisional while vessel information, insurance evidence or harbour approval is checked.	A provisional hold blocks the berth for other vessels.	Q04: Is there a maximum duration for a provisional hold? What happens if evidence is not received? A01: A provisional hold blocks the berth while checks are in progress (temporary assumption). Q24: Who is allowed to mark a request as provisional or confirmed?	P3
E07	Staff disagree how long provisional holds should block other vessels.	There is no agreed rule. The vessel operator needs to know how long a berth can be held.	See Q04	P3
E08	Equipment may be reserved with a berth or independently.	No problem noted.	Q05: Who may reserve equipment without a berth? Is there a limit per person?	Question -> register
E09	Compatibility, rated capacity, connectors, inspection status and complete accessories matter.	A compatibility error means the equipment will not suit the vessel.	Q06: Who confirms that equipment is compatible with the vessel? What happens if issued equipment is unsuitable?	P4
E10	An item can be stored at the harbour but unavailable because it is installed elsewhere, wet, damaged or awaiting inspection.	‘At the harbour’ does not mean ‘available’.	Q07: Can vessel operators see equipment that is currently unavailable?	P4
E11	Sets can be incomplete even when the main component is present.	An incomplete set is a safety risk for the vessel operator.	A02: An incomplete set is treated as unavailable (temporary assumption). Q08: May an incomplete set ever be issued?	P4
E12	Before issue, staff confirm the responsible person, intended use and current checks.	Rated capacity and inspection matter as much as physical availability.	Q09: Which checks are required, and how often must they be repeated to count as current?	P4
E13	Different equipment categories require different authorisation or supervision.	No problem noted.	Q10: Which equipment categories require which authorisation or supervision, and who grants it?	Question -> register
E14	On arrival, staff confirm the vessel, berth and service arrangements.	No problem noted.	None.	Current process
E15	A delayed vessel may overlap with the next booking.	Bookings overlap in time.	Q11: How does an operator report a changed arrival time? Is there an acceptable delay? What happens when bookings overlap?	P5
E16	An early arrival may need a temporary berth.	No problem noted.	Q12: How is temporary allocation for an early arrival determined? A03: An early vessel may wait behind a vessel still at the berth (an option only, not a fact).	P5

ID	Scenario evidence	Initial observation	Questions (Q) and assumptions (A)	→ Used in
E17	Weather, depth, works, incidents or vessel condition may require relocation, delayed entry or cancellation. Staff need to distinguish a changed berth from a rejected arrival.	Without this distinction, it is unclear what happened to a request.	Q13: Which conditions lead to relocation, delay or refusal? Who may relocate, delay or refuse an arrival?	P5
E18	Equipment may be moved between berth areas. Changes agreed by radio or telephone are not always written into the harbour log promptly.	A move may not be recorded: confusion, shortages, time spent searching.	Q14: Who must record an equipment move, and within what time?	P6
E19	Departure releases the berth only after relevant equipment and access arrangements are cleared.	The berth stays occupied until everything is cleared.	Q15: When do the berth and equipment become available again after departure?	P7
E20	Shared equipment may return damaged, incomplete or requiring inspection.	If much equipment returns unusable, there may not be enough for others.	Q16: What happens to existing bookings if there is not enough equipment?	P7
E21	Incidents and damage may involve vessel movement, harbour conditions or equipment. Staff record observations without deciding cause prematurely.	Cause should not be decided at the moment of recording.	Q17: What information is recorded for an incident? Who determines cause and responsibility? How is the information shared?	P8
E22	The coordinator needs views of provisional arrivals, berth conflicts, equipment readiness, restrictions, delayed departures and unresolved inspections.	No problem noted.	Q18: In what form does the coordinator need this information? Q25: What counts as a berth conflict: only two bookings at the same time, or also a vessel placed in a berth that doesn't suit it?	P9
E23	Navigation control, vessel maintenance and financial accounting are outside scope.	Payments, penalties, extra charges and profit are out of scope.	Q19: How is payment handled? (boundary only, not designed)	Boundary -> scope

1.2 Symptoms, problems, consequences and desired outcomes

ID	Observation or symptom	Underlying problem	Likely consequence	Desired outcome
P1	Berth requests, vessel details, equipment reservations, inspections and notices are kept in separate logs and messages [S]	Records are spread across separate logs and messages, so they are easy to lose or mix up [I]	A booking may be missed or lost, and available berths and equipment may be counted incorrectly [I]	Accurate and consistent information about requests, berths and equipment [I]
P2	An empty berth may be unsuitable for a vessel or already reserved for that time [S]	Suitability depends on length, depth, access, power, restrictions and time window [S]	A vessel may be given an unsuitable or already reserved berth [I]	Staff can tell whether a berth is suitable and free [I]

ID	Observation or symptom	Underlying problem	Likely consequence	Desired outcome
P3	Some requests are provisional while vessel information, insurance and harbour approval are checked; staff disagree how long to hold the berth [S]	There is no agreed rule for provisional holds [S]	The operator does not know how long the berth is held [S]; the berth is blocked for other vessels [I]	Hold status and duration are clear; the duration itself remains an open question [I]
P4	Equipment is stored at the harbour but unavailable; sets may be incomplete [S]	Stored does not mean complete, inspected and available [S]; rating and inspection matter as much as physical availability [S]	Incomplete, uninspected or incompatible equipment may be issued, which affects user safety [I]	Each item shows its current status and, if unavailable, the reason (installed elsewhere, wet, damaged, awaiting inspection, incomplete set); only ready items are issued [I + S]
P5	A delayed vessel overlaps the next booking; an early arrival needs a temporary berth; weather, depth or works cause relocation, delay or cancellation; staff must distinguish a changed berth from a rejected arrival [S]	There are no written rules for when to change a berth and when to refuse an arrival [I]	Bookings overlap in time [I]	It is clear what happened to each arrival (changed berth, delay or refusal) and why; the rules themselves are a stakeholder question [I]
P6	Equipment is moved between berth areas; changes agreed by radio or telephone are not written into the log promptly [S]	Moves may not be recorded [I]	Confusion, equipment shortages, time spent searching [I]	The location of equipment is always known [I]
P7	A berth is released only after equipment and access are cleared; equipment returns damaged or incomplete [S]	It is unclear when the berth and equipment become available again [S]	If much equipment returns unusable, other vessels may not have enough [I]	Staff know when a berth and equipment become available again [I]
P8	Incidents may involve the vessel, harbour conditions or equipment; observations must be recorded without deciding cause [S]	How incidents should be recorded and shared is not defined [S]	Without records, the cause cannot be established later [I]	Staff record observations that allow the cause to be established later [I]
P9	The coordinator needs views of provisional arrivals, conflicts, equipment readiness, restrictions, delays and unresolved inspections [S]	The information is held in separate logs and messages [S]	The coordinator struggles to get an overall picture [I]	The coordinator can see these six areas; the form is not yet known [I]

Excluded: profit and lost revenue (financial accounting is outside scope in the scenario); sensors, AI tracking and a fixed number of items per berth (these are proposed solutions, not problems); the format of the coordinator's views, such as charts or tables (also a solution, see Session 3 Laboratory, Guided exercise 2).

1.3 Problem statement

Kerry Harbour Services Cooperative coordinates short-term berth use and shared harbour equipment for fishing, leisure and community vessels. Berth requests, vessel details, equipment reservations, inspections and notices are kept in separate logs and messages. With so much to record, staff can easily miss who booked what, lose records or mix information up, so bookings may be lost and available berths and equipment miscounted. An empty berth is not always suitable for a vessel, and staff have not agreed how long a provisional hold should block a berth. Equipment may be at the harbour but not ready to issue, which is a safety issue. Changes agreed by radio may not be written down, so equipment has to be searched for. Delayed vessels can overlap with the next booking, and it is not always clear whether a vessel was moved or refused entry. If much equipment returns damaged, other vessels may not have enough. How incidents are recorded and shared is undefined, and the coordinator struggles to get an overall picture. These issues affect the harbour coordinator, harbour assistants, the equipment officer, the safety officer and vessel operators. As a result, staff spend more time on administration and make recording errors, while operators do not know how long their berth will be held.

2. Purpose and scope

2.1 Purpose

The proposed Harbour Berth and Equipment Management System will support berth requests and provisional holds, checking which berths are suitable and free, allocating berths, booking, issuing and returning shared equipment, keeping track of equipment status and location, recording arrivals, departures and incidents, and giving the harbour coordinator an overview of what is happening.

2.2 Scope

In scope for the initial release	Out of scope for the initial release
Berth requests, vessel details and operator contacts	Navigation control
Provisional holds	Vessel maintenance
Berth suitability and availability, including restrictions and works	Financial accounting, payments and charges
Berth allocation and changes (relocation, delay, refusal)	Deciding harbour rules, e.g. how long a hold lasts or when to refuse an arrival. Staff make these decisions; the system only applies and records them
Equipment reservations, with or without a berth	Tracking equipment automatically with sensors
Equipment status, inspections and location	Managing staff and shift rotas
Equipment authorisation and supervision requirements	A full customer database
Equipment issue and return, including the responsible person	Improving harbour facilities
Arrivals, departures and berth release	Water-use rights and legal contracts
Incident and damage observations	Lifeguard posts, lighting and CCTV
Harbour notices and operational restrictions	Providing crew for vessels
Coordinator views	

2.3 Dependencies, constraints and assumptions

The system depends on some things it cannot control. Insurance evidence comes from the vessel operator or insurer, not from the system (D01). Harbour approval also comes from someone outside the system, and the scenario does not say who (D02, Q02). Changes agreed by radio or phone only get into the system if a staff member enters them (D03). D01 and D02 come from the scenario; D03 is based on the scenario and my own view.

The main constraint from the scenario is that the system records what staff observe about an incident but does not decide the cause (C01).

Some points are assumptions, not facts. Assumptions A01 to A03 and A07 are in Section 1.1. The full list of rules, assumptions, constraints, risks and questions is in Section 5.

3. Stakeholders and actors

3.1 Stakeholder analysis

Direct users People who will use the system themselves.

Stakeholder	Goals	Concerns	Influence	Expected interaction
Harbour coordinator quoted in the scenario	Before a vessel is placed, check that the berth actually suits it. Only release a berth after the equipment is returned and access is closed.	A berth is free but does not suit the vessel, so it stands empty and vessel operators are delayed.	High	Confirms requests, allocates berths, records harbour notices, reserves equipment and uses the overview.
Equipment officer quoted in the scenario	Equipment is checked properly for faults before it can be booked. It is clear which equipment is ready and which should not be used until it is fixed.	Broken equipment could be given out, which puts people's safety at risk.	Medium	Records equipment status, inspections and location; checks equipment when it comes back.
Harbour assistant quoted in the scenario	Berth changes are agreed and recorded so there is no confusion.	A berth change that reaches the log late could affect the actual move, so a vessel ends up at a faulty or unsuitable berth.	Medium	Records berth changes agreed by radio, arrivals, departures and what they see during incidents.
System administrator not mentioned in the scenario	The system is available and secure, and everyone has the right access.	System failures; someone getting access they should not have.	High	Manages user accounts, access and the running of the system.

Indirect stakeholders Affected by the system or set the rules, but do not use it day to day.

Stakeholder	Goals	Concerns	Influence	Expected interaction
Vessel operator quoted in the scenario	If their arrival at the booked berth changes because of weather or anything else, to have an idea how long the berth will be held for them. For someone to contact them or let them know how long the berth can be held.	When they arrive, the berth is already taken by another vessel and there is nowhere to moor.	Medium	Sends the berth request and vessel details, finds out the booking status and any changes (directly or through staff, not clear yet).
Safety officer quoted in the scenario	Equipment is checked for its load rating and everything else just as carefully as whether it is physically there. Different equipment categories follow the right authorisation and supervision.	Equipment that cannot take the load or has not been inspected is given out, and someone gets hurt or a vessel is damaged.	High	Does not use the system directly. Staff check load ratings, inspection status and authorisation in the system (Section 3.2).

Stakeholder	Goals	Concerns	Influence	Expected interaction
Cooperative management not mentioned in the scenario, my assumption	The system works properly and the harbour runs without confusion. Berths and equipment are not left unused. Staff have clear rules (e.g. how long to hold a booking). Can see the overall picture.	Accidents or injuries, as they are responsible. Making decisions based on wrong or incomplete information. Staff disagreements (e.g. about hold times) never get settled. Vessel operators get fed up with berth mix-ups and go to another harbour.	High	Looks at the overall picture; sets harbour rules outside the system.
Data-protection officer not mentioned in the scenario	Personal data (vessel operators' contact details) is handled legally.	Personal data leaks, data kept longer than it should be.	High	Does not use the system every day; checks how personal data is stored.

External stakeholders Outside the cooperative.

Stakeholder	Goals	Concerns	Influence	Expected interaction
Insurer implied in the scenario ("insurance evidence")	Accurate and complete incident records, so the cause and the cost of the damage can be worked out.	Records are wrong or incomplete, so it is not clear whether the insurance covers the damage.	Low	Does not use the system directly; insurance evidence comes through the vessel operator.
Harbour approval authority implied in the scenario ("harbour approval"); who exactly is unknown, Q02	Vessel operators can enter all the details needed, the details are checked, and bookings can be confirmed faster. Can see which bookings are waiting for approval.	Approving a booking for a vessel that is not fully equipped or safe because the details were wrong or missing.	High	Gives approval for bookings; not clear if this happens through the system.

3.2 Actors and goals

Actor or stakeholder	Type	Goal or interest	Direct interaction?
Harbour coordinator	Primary actor	Check which berths fit and are free, confirm requests and choose berths, record harbour notices, reserve equipment and see the overall picture.	Yes
Equipment officer	Primary actor	Save the status, checks and location of equipment, give it out and take it back after use.	Yes
Harbour assistant	Primary actor	Save berth requests and changes from operators, berth changes agreed by radio, arrivals, departures and incident notes.	Yes

Actor or stakeholder	Type	Goal or interest	Direct interaction?
System administrator	Primary actor	Manage accounts and access.	Yes
Vessel operator	Offstage stakeholder	Get a berth for the vessel and know how long it will be held if the arrival time changes.	No (harbour staff do it for them, open question Q23)
Safety officer	Offstage stakeholder	Make sure equipment is checked and can take the load before it is given out. Staff do the checking in the system.	No
Harbour approval authority	Offstage stakeholder	Approve requests that are still provisional.	No (it is not clear if they use the system, Q02)
Insurer	Offstage stakeholder	Insurance evidence is checked before a request is confirmed.	No (the evidence comes through the operator)
Cooperative management	Offstage stakeholder	Set the rules and decide about the system.	No
Data-protection officer	Offstage stakeholder	Make sure personal data, like operator contacts, is handled properly.	No

4. Requirements

4.1 Functional requirements

ID	Requirement	Source	Priority	How to check
FR01	The system shall let staff save a berth request with the vessel's details, the arrival and departure time, why the vessel is coming, the services it needs, and the operator's contact.	E04	Must	Demonstration: save a request and check that all the details are there.
FR02	The system shall let staff change a berth request when the operator says plans have changed.	E04, Q22	Must	Demonstration: change the arrival time and check that the new time is shown.
FR03	The system shall let staff mark a request as provisional, note which check is still waiting (vessel information, insurance or harbour approval), and show that the berth is being held.	E06, BR02, A01	Must	Demonstration: mark a request as provisional and check that the berth shows as held.
FR04	The system shall show which berths fit the vessel and are free at the time it wants.	E05, Q03	Must	Test: ask for a vessel and compare the berths shown with the berth details.
FR05	The system shall let staff save harbour notices, restrictions and works that affect a berth.	E03, E05	Should	Demonstration: add a restriction and check that the berth shows as affected.
FR06	The system shall let staff choose a berth for a request and save the choice.	E04, E05	Must	Demonstration: choose a berth and check that it appears on the request.
FR07	The system shall let staff record a berth change as a move to another berth, a delay, or a refusal.	E17, Q13	Must	Demonstration: record each of the three and check that they are shown as different things.
FR08	The system shall let staff book equipment together with a berth or on its own.	E08, BR01	Must	Demonstration: make one booking with a berth and one without.
FR09	The system shall keep the status, checks and location of each piece of equipment.	E09, E10, E18	Must	Demonstration: update a piece of equipment and check that the new details are saved.
FR10	The system shall treat equipment as unavailable if it is incomplete or has not been checked.	E10, E11, BR07, A02	Must	Test: mark an item as incomplete and check that it shows as unavailable.
FR11	The system shall ask staff to confirm who is responsible, what the equipment is for, and that its checks are up to date before it is given out.	E12, BR03	Must	Test: try to give out equipment without confirming and check that the system stops it.
FR12	The system shall show what permission or supervision each type of equipment needs.	E13, BR04, Q10	Could	Inspection: look at an equipment type and check that this information is shown.
FR13	The system shall record when equipment is given out and returned, who is responsible, and what condition it is in when it comes back.	E12, E20	Must	Demonstration: give out and return an item and check the record.

ID	Requirement	Source	Priority	How to check
FR14	The system shall let staff record a vessel's arrival (checking the vessel, berth and services) and its departure.	E14, BR05	Must	Demonstration: record an arrival and a departure.
FR15	The system shall only free a berth after the equipment and access arrangements are cleared.	E19, BR06, Q15	Must	Test: let a vessel depart with equipment not cleared and check that the berth stays occupied.
FR16	The system shall let staff save notes about incidents and damage.	E20, E21, Q17	Should	Demonstration: save an incident note and check that it is stored.
FR17	The system shall give the coordinator one overview of provisional arrivals, berth conflicts, equipment readiness and location, restrictions, delayed departures and unresolved inspections.	Coordinator quote, E22, E18	Must	Demonstration: open the overview and check that all six things are shown.
FR18	Tracking equipment automatically with sensors (not in this release).	Section 2, out of scope	Won't	Not checked: not built in this release.
FR19	Payments and charges (not in this release).	Section 2, out of scope	Won't	Not checked: not built in this release.
FR20	Managing staff and shift rotas (not in this release).	Section 2, out of scope	Won't	Not checked: not built in this release.
FR21	A full customer database with history and profiles (not in this release). Operator contacts and vessel details are still saved in FR01.	Section 2, out of scope	Won't	Not checked: not built in this release.
FR22	The system shall let staff confirm a berth request when no check is missing.	E06, Q02, Q24	Must	Demonstration: confirm a request with no missing check, then try with a missing check and check that it stays provisional.

4.2 Non-functional requirements

ID	Requirement	Quality	Source	Priority	How to check
NFR01	Only harbour staff who are allowed to shall be able to see the operator's contact details or change requests, equipment records and berth choices.	Security	Section 3 (data-protection officer), FR01	Must	Inspection
NFR02	Nothing that staff have saved (requests, holds, equipment records, equipment moves) shall be lost when the system is restarted once.	Reliability	E02, E18, P1	Must	Test
NFR03	Every berth change, every equipment issue or return, and every incident note shall show who saved it and when.	Traceability	E18, E21, BR03	Must	Inspection

ID	Requirement	Quality	Source	Priority	How to check
NFR04	The list of suitable berths (FR04) shall show up within 3 seconds for at least 95 out of 100 requests when the system is used normally.	Performance	E03, E05	Should	Test
NFR05	The system shall work during the harbour's working hours. The only exception is maintenance that staff were told about at least 24 hours before.	Availability	E03	Should	Analysis
NFR06	A trained staff member shall be able to save an equipment move in under 60 seconds.	Usability	E18, Q14	Could	Test

5. Rules and uncertainty register

5.1 Business rules

ID	Business rule	Source
BR01	Equipment can be booked together with a berth or on its own.	E08
BR02	A berth request stays provisional while vessel information, insurance evidence or harbour approval is being checked.	E06
BR03	Before equipment is given out, staff confirm who is responsible for it, what it will be used for, and that its checks are up to date.	E12
BR04	Different equipment categories need different authorisation or supervision. The scenario does not say which ones (Q10).	E13
BR05	When a vessel arrives, staff confirm the vessel, the berth and the services agreed.	E14
BR06	A berth is only released after the related equipment and access arrangements are cleared.	E19; coordinator quote
BR07	Equipment does not count as available just because it is stored at the harbour. It also has to be complete, inspected and suitable for the load.	E09, E10; equipment officer and safety officer quotes

5.2 Risks

These risks could stop the system from working as planned. Each risk is followed by a response.

R01. Old records may be incomplete, wrong or duplicated. If they are moved into the new system, the mistakes move with them. Response: check the old logs and messages before moving them.

R02. Staff may keep agreeing changes by radio and forget to enter them, so the system shows out-of-date information again. Response: find out who has to enter changes and how quickly (Q14).

R03. Many harbour rules are not agreed yet (e.g. how long to hold a booking, who can refuse an arrival). If the answers take a long time, some requirements cannot be finished. Response: ask the coordinator and management the open questions as early as possible.

R04. Staff already disagree about how long to hold a booking. If a rule is chosen that they do not like, they may not want to use the system. Response: talk the rules through with staff before they are agreed.

R05. Equipment status may not be updated on time (e.g. it has dried or passed its inspection), so the system shows the wrong readiness. Response: find out who is responsible for updating equipment status.

5.3 Assumptions

ID	Assumption	Where it comes from
A01	A provisional hold blocks the berth while checks are in progress.	My own view (Section 1.1, E06)
A02	An incomplete equipment set is treated as unavailable.	My own view (Section 1.1, E11)
A03	An early vessel may wait behind a vessel that is still at the berth. This is only an option, not a fact.	My own view (Section 1.1, E16)
A04	The cooperative has a management group that agrees the harbour rules.	Section 3, not mentioned in the scenario
A05	There will be a system administrator who manages user access.	Section 3, based on the Session 3 lab example
A06	Vessel operators' contact details are personal data, and someone is responsible for protecting them.	Section 3, based on the Session 3 lab example
A07	Harbour staff enter berth requests and changes for the vessel operators. The operators do not use the system themselves. This is only an assumption, not a fact (see Q23).	My own view (Section 1.1, E04)

5.4 Constraints

The system has to work within these limits.

C01. The system records what staff observe about an incident but does not decide the cause. This comes from the scenario and is also in Section 2.3.

C02. Navigation control, vessel maintenance and financial accounting are not part of the system. This comes from the scenario and is also in Section 2.2.

C03. Personal data (vessel operators' contact details) must be handled in line with data-protection law. This is not stated in the scenario; it is based on assumption A06.

5.5 Stakeholder clarification questions

ID	Question	Why the answer matters	Affects
Q02	Which vessel details and evidence are needed before a booking is confirmed?	Without this it is unclear when a provisional booking becomes confirmed.	P3, BR02
Q04	How long should a provisional hold last? What happens if the documents do not arrive?	This decides how long a berth is blocked and what to tell the vessel operator.	P3, BR02, A01
Q03	How is it decided whether a berth suits a vessel?	Without clear rules, suitability cannot be checked.	P2
Q12	Where should a vessel that arrives early be placed?	It is unclear what to do with early arrivals.	P5, A03
Q10	Which equipment needs authorisation or supervision, and who gives it?	Without this it is not possible to decide if equipment can be given out.	P4, BR04

ID	Question	Why the answer matters	Affects
Q09	Which equipment checks are needed, and how often?	It is unclear when a check counts as up to date.	P4, BR03
Q13	When is a vessel moved, delayed or refused, and who decides?	Needed to tell a changed berth from a refused arrival, and to know who has the right to decide.	P5
Q22	How far in advance must an operator report a change, and how?	This decides whether the harbour has time to give the berth to someone else.	P3, P5
Q14	Who must record equipment moves, and how quickly?	Otherwise the log will fall behind again.	P6, R02
Q15	When do the berth and equipment become available again after departure?	This decides when the berth can be given to the next vessel.	P7, BR06
Q17	What is recorded for an incident, who decides the cause, and how is it shared?	So that records are complete and reach the people who need them.	P8, C01
Q23	Do vessel operators use the system themselves, or only through harbour staff?	It decides if operators are users of the system, and if we need accounts and access rules for them.	A07, FR01, FR02, FR03, NFR01
Q24	Who is allowed to mark a request as provisional or confirmed?	This decides who can hold a berth and who can confirm it.	P3, FR03, FR22
Q25	What counts as a berth conflict: only two bookings at the same time, or also a vessel placed in a berth that doesn’t suit it?	This decides what the coordinator’s overview must show as a conflict.	P5, P9, FR17
Q26	Which staff roles are allowed to see operator contact details?	This decides who gets access to personal data.	NFR01
Q27	Which staff role records berth requests and their changes, and which role gives out equipment?	This decides which actor each of these use cases belongs to.	FR01, FR02, FR11, FR13

The other questions (Q01, Q05 to Q08, Q11, Q16 and Q18 to Q21) are in Section 1.1.

6. Use-case model

6.1 Use-case catalogue

ID	Use case	Primary actor	Meaningful outcome	Source requirement
UC01	Record Berth Request	Harbour assistant	A berth request with the vessel details, time and services is saved	FR01
UC02	Update Berth Request	Harbour assistant	The request shows the latest plan from the operator	FR02
UC03	Confirm Berth Request	Harbour coordinator	The request is confirmed, or it is marked provisional with the missing check shown and the berth held	FR03, FR22
UC04	Allocate Berth	Harbour coordinator	A suitable and free berth is given to the request	FR04, FR06
UC05	Record Berth Change	Harbour assistant	A move, delay or refusal is saved in the log	FR07
UC06	Record Harbour Notice	Harbour coordinator	A notice, restriction or works affecting a berth is saved	FR05
UC07	Reserve Equipment	Harbour coordinator	Equipment is kept for a vessel, with a berth or on its own	FR08
UC08	Update Equipment Status	Equipment officer	The status, checks and location of the equipment are saved, and unready equipment shows as unavailable	FR09, FR10
UC09	Issue Equipment	Equipment officer	Equipment goes out only after the responsible person, use and checks are confirmed	FR11, FR12, FR13
UC10	Return Equipment	Equipment officer	The return and the condition of the equipment are saved	FR13
UC11	Record Vessel Arrival	Harbour assistant	The arrival is saved after the vessel, berth and services are checked	FR14
UC12	Record Vessel Departure	Harbour assistant	The departure is saved, and the berth is freed only after equipment and access are cleared	FR14, FR15
UC13	Record Incident	Harbour assistant	An incident or damage note is saved without deciding the cause	FR16
UC14	View Harbour Overview	Harbour coordinator	Provisional arrivals, berth conflicts, equipment readiness and location, restrictions, delayed departures and unresolved inspections are seen in one place	FR17
UC15	Grant Staff Access	System administrator	Only staff who are allowed to can see contact details and change records	NFR01

Note on primary actors

The scenario does not say which staff role records berth requests, confirms them, records harbour notices, reserves equipment or issues equipment. The primary actors for UC01, UC02, UC03, UC06, UC07 and UC09 are my own decision, not a fact from the scenario. The open questions are Q27 (requests and issuing equipment), Q24 (confirming requests), Q20 (harbour notices and restrictions) and Q05 (reserving equipment), see Sections 1.1 and 5. The other primary actors follow the stakeholder quotes and goals in Section 3.

Note on FR03, FR04, FR10, FR15 and FR22

Confirm Berth Request has two outcomes: the request is confirmed (FR22), or it stays provisional while a check is missing (FR03). This follows Request Activity Place in the Session 4 sample answer, which records a confirmed place or a waiting-list outcome. Who may confirm a request is still open (Q24).

FR10 (unready equipment shows as unavailable) and FR15 (a berth is freed only after equipment and access are cleared) could be treated as rules within Update Equipment Status and Record Vessel Departure rather than separate use cases. FR04 (show suitable and free berths) is a step of Allocate Berth rather than a separate goal. Not every functional requirement must become a separate use case.

Requirement coverage

Every Must, Should and Could functional requirement (FR01 to FR17 and FR22) and NFR01 appears in at least one use case. FR18 to FR21 are Won’t items for this release, so they have no use case. The other non-functional requirements (NFR02 to NFR06) describe qualities of the system, not actor goals, so they stay in the requirements catalogue (Section 4).

6.2 Use-case diagram

Source file: diagrams/use-case-model.puml.

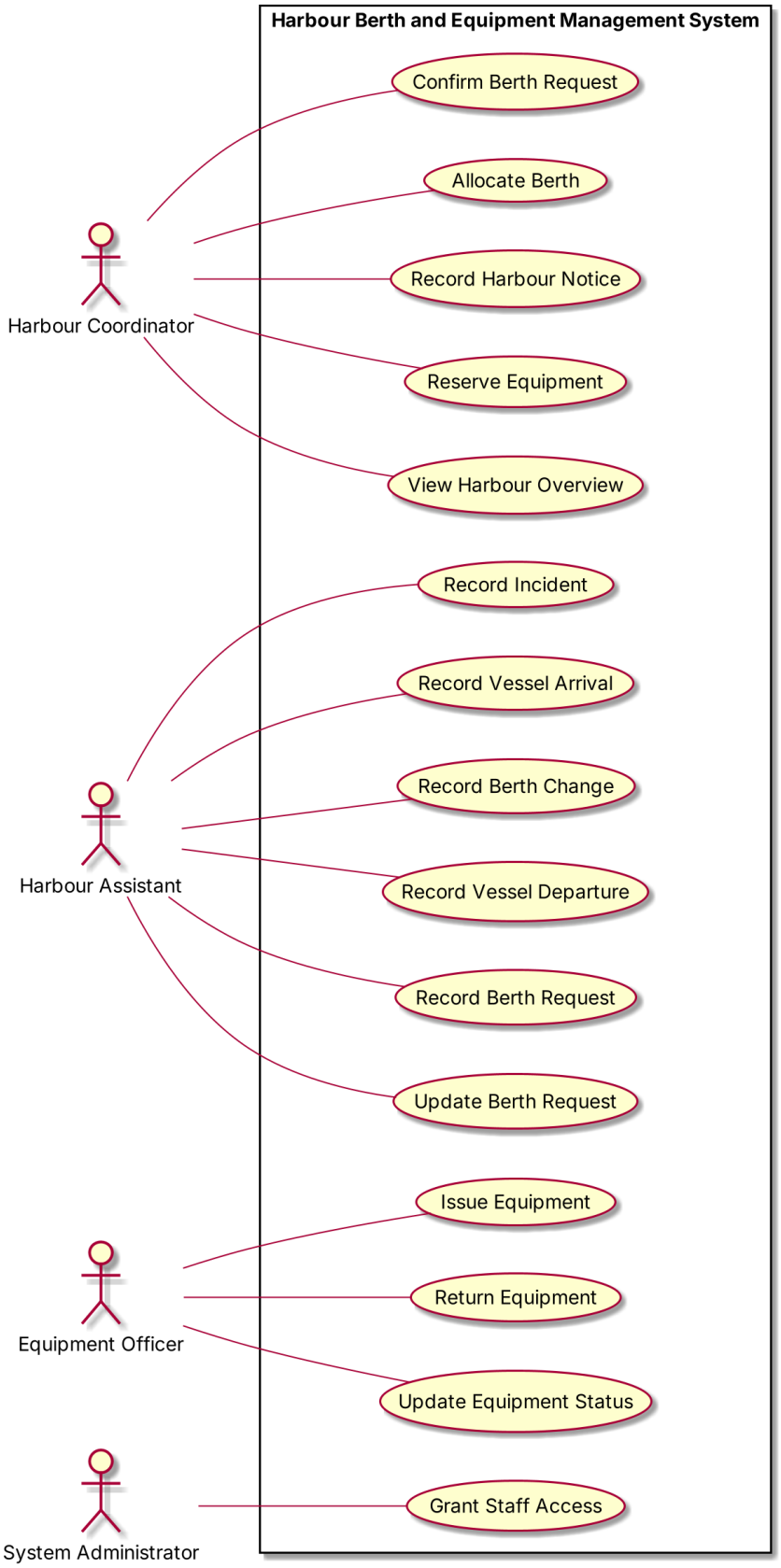


Diagram justification

- All actors appear outside the system boundary.
- Use cases appear inside the named system boundary.
- The use-case names use verb-object wording.
- Associations show participation rather than message order.
- Each use case is linked to its primary actor from the catalogue (6.1), so not every actor is connected to every use case.
- No supporting actor is drawn because the scenario does not mention an outside system that this system calls (Section 3.2).
- Vessel operators, the safety officer, the insurer and the other offstage stakeholders are not drawn because they do not use the system directly (Section 3.2, A07).
- No database, Java class, screen, field or controller appears in the diagram.
- No include or extend relationship is needed at this stage.

6.3 Detailed use case: UC04 Allocate Berth

6.3.1 Header

Field	Content
ID and name	UC04 Allocate Berth
Scope	Harbour Berth and Equipment Management System
Level	User goal
Primary actor	Harbour coordinator
Preconditions	The berth request is recorded with the vessel details, arrival and departure time and services needed (UC01). The length, depth, access and power of each berth are known to the system (assumption: the scenario does not say how berth details are kept).
Trigger	The coordinator decides to give a berth to a recorded berth request.
Minimal guarantee	No unsuitable berth is allocated, and no berth is allocated twice for the same time. When the goal cannot be completed, the reason is shown.
Success guarantee	The request has one suitable berth for its arrival and departure time, and the berth is shown as allocated for that time.
Open issue	The exact suitability rule is not agreed (Q03). How a temporary berth is given to an early arrival is not established (Q12). That a provisional hold blocks the berth is an assumption (A01). If the request is still provisional, the berth is shown as held rather than allocated (FR03, UC03).

6.3.2 Main Success Scenario

Step	Actor intention or system responsibility
1	The coordinator selects the berth request to be allocated.

Step	Actor intention or system responsibility
2	The system shows the vessel details, the arrival and departure time and the services needed.
3	The system finds the berths that fit the vessel's length and depth, have suitable access and offer the services needed, such as power.
4	The system confirms that these berths are not allocated or held for any part of the requested time.
5	The system confirms that no harbour notice, restriction or works affects these berths at that time.
6	The system shows the suitable and free berths.
7	The coordinator chooses one berth.
8	The coordinator confirms the allocation.
9	The system records the berth on the request for the requested time.
10	The system shows the berth as allocated for that time.

6.3.3 Extensions

Ref.	Condition	Response and destination
3a	No berth fits the vessel or offers the services needed.	Explain that no suitable berth exists; allocate no berth; end.
4a	All suitable berths are allocated or held for part of the requested time.	Show the berths and the time that overlaps. If the operator agrees a revised time, the request is updated (UC02) and the flow resumes at step 3; otherwise allocate no berth; end.
5a	A restriction or works affects a suitable berth for part of the requested time.	Show the restriction; leave that berth out; resume at step 6 with the other berths. If no berth is left, continue as 3a.
8a	The coordinator cancels before confirming.	Allocate no berth; the request stays without a berth; end.

6.3.4 Quick classification check

- 3a is a failure extension.
- 4a and 5a are recoverable extensions.
- 8a is a cancellation extension.

7. Gherkin examples

7.1 Selected examples

ID	Scenario	Source in UC04	Type	Links
GH01	Coordinator allocates a suitable free berth	Main Success Scenario	Success	FR04, FR06, US03
GH02	No berth is long enough for the vessel	Extension 3a	Failure	FR04, US03
GH03	A berth that is taken for part of the time is not shown as free	Extension 4a	Alternative	FR04, US03
GH04	A berth closed for works is not shown as suitable	Extension 5a	Rule	FR05, US03

There is no number-based boundary example. The scenario does not say if a booking that ends at 18:00 overlaps with one that starts at 18:00, so writing that example would invent a rule. GH04 is used as the rule example instead.

The full scenarios are in the feature file features/allocate-berth.feature in the repository.

8. Candidate concepts and glossary

8.1 Candidate extraction

Candidate term	Evidence source	Initial question or observation
harbour coordinator, harbour assistant, equipment officer, staff, vessel operator, safety officer, responsible person, system administrator	Stakeholder evidence; Shared equipment; UC15	Roles. Check who uses the system (Section 3.2).
issue, return, move, release, clear, check, block, reserve, change, allocate, confirm	All sections; FR06, FR22; UC03, UC04	Verbs. Some may show a record, like borrow shows Loan.
radio, telephone, log, harbour log, message, plan, navigation control, vessel maintenance, financial accounting	Organisation; Arrival; Departure	Current way of working, or named as outside scope.
Kerry Harbour Services Cooperative, harbour, berth use	Organisation	The organisation itself and very general words.
berth, temporary berth, berth area, length, depth, access, power	Vessels, berth requests and suitability; Arrival	A berth has its own details. The other words may describe it.
vessel, vessel type (fishing, leisure, community), dimensions, vessel details, vessel information, vessel condition, vessel movement	Organisation; Vessels; Arrival; Departure	A vessel has its own details. Several words mean the same data.
request, booking, provisional hold, equipment reservation, arrival and departure window, time window, purpose, contact, required services, service arrangement	Vessels; Shared equipment; Arrival	A request has details and a status. Booking may be the same thing.
insurance evidence, harbour approval	Vessels	Checks that keep a request provisional.
equipment, item, set, main component, accessories, equipment category, mobile fender, shore-power lead, pump, lifting accessory, safety barrier, compatibility, rated capacity, connector, equipment condition (wet, damaged), equipment readiness, installed location	Organisation; Shared equipment	A general word, one item and a type may be different things. Many words describe an item.
inspection, inspection status, unresolved inspection	Shared equipment; Departure	Has a date and result and can stay open.
intended use, authorisation, supervision	Shared equipment	Things confirmed before equipment is given out.
arrival, early arrival, departure, delay, delayed entry, delayed departure, access arrangement	Arrival; Departure	Times and states of one visit.
vessel change, relocation, cancellation, changed berth, rejected arrival, overlap, berth conflict	Organisation; Arrival; Departure	Changes must be told apart. Conflicts may be worked out from bookings.
operational restriction, harbour notice, works, weather, harbour conditions	Organisation; Vessels; Arrival	Notices affect a berth for a time.
incident, damage, observation, cause	Departure	Observations are recorded without deciding the cause.

Candidate term	Evidence source	Initial question or observation
view	Departure	The coordinator's views.

8.2 Classification

Candidate	Classification	Reason
harbour coordinator, harbour assistant, equipment officer, staff, system administrator	Actor	They use the system directly (Section 3.2).
vessel operator, safety officer	Stakeholder, not actor	Offstage stakeholders (Section 3.2, A07).
issue, return, move, release, clear, check, block, reserve, change, allocate, confirm	Operation or event	Actions. Issue and return show a record of giving out equipment.
radio, telephone, log, harbour log, message	Out of scope (current process)	The current way of recording, not domain meaning.
navigation control, vessel maintenance, financial accounting	Out of scope	The scenario says they are outside scope.
Kerry Harbour Services Cooperative	System boundary	Names the organisation, like CampusHub in the lab.
harbour, plan, berth use	Too general	No details of their own.
berth, temporary berth, vessel, request, booking, equipment reservation, equipment, item, equipment category, inspection, incident, harbour notice, operational restriction, works, relocation / delay / delayed entry / cancellation / changed berth / rejected arrival / vessel change	Conceptual class	The harbour must remember their details, tell them apart, and their status changes.
length, depth, access, power, berth area, installed location	Attribute	Simple values that describe a berth or an item.
vessel type, dimensions, vessel condition, vessel details, vessel information, vessel movement	Attribute or synonym	Simple values about a vessel; details and information mean the same data.
provisional hold, arrival and departure window, time window, purpose, contact, required services, service arrangement, insurance evidence, harbour approval, arrival, departure, access arrangement	Attribute	Simple values or a status of a request.
set, main component, accessories, compatibility, rated capacity, connector, equipment condition, equipment readiness, inspection status, unresolved inspection	Attribute	Simple values that describe an item or its check.
mobile fender, shore-power lead, pump, lifting accessory, safety barrier	Value (example)	Examples of equipment categories.

Candidate	Classification	Reason
responsible person, intended use, authorisation, supervision	Attribute	Values on a record of giving out equipment or on a category.
damage, observation, cause, weather, harbour conditions	Attribute	Values that describe an incident, a notice or a change.
overlap, berth conflict, delayed departure, early arrival	Derived information	Worked out from requests, berths and times.
view	Generated output	Produced from other records.

8.3 Refinement decisions

Candidate	Decision	Final term or destination	Justification
berth, temporary berth	Keep / Merge	Berth	Has length, depth, access, power and a changing status. A temporary berth is a normal berth used for a short time (Q12).
vessel, vessel details, vessel information	Keep / Merge	Vessel	The same data about one vessel. A vessel can come more than once.
request, booking	Merge	BerthRequest	The same thing. The status shows if it is provisional or confirmed.
allocate	Remove from class list	Link between BerthRequest and Berth	The verb shows that a request is given one berth (FR06, UC04). This is a link between two concepts, not a separate concept. Changes to the berth are kept in BerthChange.
provisional hold	Convert	Status of BerthRequest	Not separate data. How long it lasts is still open (Q04).
insurance evidence, harbour approval	Convert	Missing checks on BerthRequest	Shows which check is still waiting (FR03). May become a class after Q02.
arrival and departure window, purpose, contact, required services, service arrangement	Convert	Attributes of BerthRequest	“A request includes…”. Service arrangement means the same as required services. The operator's contact is kept here because the operator is not a concept.
arrival, departure, access arrangement	Convert	Attributes of BerthRequest	Actual times and clearance (FR14, FR15).
relocation, delay, delayed entry, cancellation, changed berth, rejected arrival, vessel change	Merge	BerthChange	One kind of record with move, delay or refusal (FR07). One request can have several changes.
harbour notice, operational restriction, works	Merge	HarbourNotice	Affects a berth for a period (FR05).
equipment, item	Merge	EquipmentItem	One physical piece with its own status, location and checks, like BookCopy in the lab.
equipment category, mobile fender, shore-power lead, pump, lifting accessory, safety barrier	Keep / Convert	EquipmentCategory	A type with its own authorisation rules, like Book in the lab. The listed kinds are examples of categories.

Candidate	Decision	Final term or destination	Justification
set, main component, accessories	Convert	Completeness of EquipmentItem	An incomplete set is unavailable (A02).
compatibility, rated capacity, connector, condition, readiness, location, berth area	Convert	Attributes of EquipmentItem	Simple values.
equipment reservation	Keep	EquipmentReservation	Can exist with a berth request or on its own (FR08).
issue, return	Keep (shown by the verbs)	EquipmentIssue	Like borrow shows Loan. Keeps the responsible person, intended use and condition on return (FR11, FR13).
responsible person, intended use	Convert	Attributes of EquipmentIssue	Simple values for one issue.
authorisation, supervision	Convert	Attributes of EquipmentCategory	Depend on the category (Q10).
inspection	Keep	Inspection	Has a date and result and can stay unresolved (E22).
incident, damage	Keep / Merge	IncidentReport	Observations are kept, the cause is not decided yet (FR16, Q17).
vessel operator	Remove for now	Offstage stakeholder (Section 3)	Offstage (A07). Revisit if Q23 is answered yes.
safety officer	Remove	Offstage stakeholder (Section 3)	Nothing about this role must be remembered.
harbour coordinator, harbour assistant, equipment officer, staff, system administrator	Remove from class list	Actors	They use the system, but no extra data about them is needed.
overlap, berth conflict, delayed departure, early arrival	Remove	Derived information	Worked out from BerthRequest, Berth and the actual times, like activity demand in the lab.
view	Remove	Generated output	Like ParticipationSummary in the lab.
radio, telephone, logs, messages	Remove	Current process	The current way of recording.
navigation control, vessel maintenance, financial accounting	Remove	Out of scope	The scenario says they are outside scope.
Kerry Harbour Services Cooperative, harbour, plan, berth use	Remove	System boundary / too general	No details of their own.
move, release, clear, check, block, reserve, change, confirm	Remove from class list	Operations	They change the status of other concepts.

Terms requiring clarification

Vessel operator

The operator is an offstage stakeholder (A07), so the term is removed for now and the contact is kept on BerthRequest. It becomes a concept only if operators use the system themselves (Q23).

Provisional hold

A hold is kept as a status of BerthRequest. How long a hold should last is not agreed (Q04).

Insurance evidence and harbour approval

They are kept as missing checks on BerthRequest. Which evidence is needed before confirmation is not agreed (Q02).

Temporary berth

It is merged into Berth. How a temporary berth is chosen for an early arrival is not agreed (Q12).

Authorisation and supervision

They are kept on EquipmentCategory. Which categories need them is not agreed (Q10).

Cause of an incident

It is not decided when the report is made. Who decides the cause and how it is shared is not agreed (Q17).

8.4 Domain glossary

Concept	Concise domain definition	Related concepts	Evidence
Berth	A place in the harbour where a vessel can stay for a short time, described by its length, depth, access and power.	Vessel, BerthRequest, HarbourNotice	“Berths differ in length, depth, access, power…”; FR04, FR06
Vessel	A fishing, leisure or community boat that needs a berth at the harbour, described by its size, type and condition.	BerthRequest, IncidentReport	Organisation and current service; FR01
BerthRequest	A request for a berth for one vessel for a time window, which can be provisional or confirmed.	Vessel, Berth, BerthChange, EquipmentReservation	“A request includes…”; “Some requests remain provisional…”; FR01 to FR03, FR22
BerthChange	A record that a berth request was moved to another berth, delayed or refused.	BerthRequest, Berth	“Staff need to distinguish a changed berth from a rejected arrival”; FR07
HarbourNotice	A notice about a restriction or works that stops a berth being used for a period.	Berth	“harbour notices”; “operational restrictions”; FR05
EquipmentCategory	A type of shared harbour equipment, such as fenders, shore-power leads or pumps, with its own authorisation or supervision needs.	EquipmentItem, EquipmentReservation	“Different equipment categories require different authorisation or supervision”; FR12

Concept	Concise domain definition	Related concepts	Evidence
EquipmentItem	One piece of shared harbour equipment with its own condition, location and checks.	EquipmentCategory, Inspection, EquipmentIssue, IncidentReport	“An item can be stored at the harbour but unavailable…”; FR09, FR10
EquipmentReservation	A booking of equipment for a period, either with a berth request or on its own.	EquipmentCategory, BerthRequest	“Equipment may be reserved with a berth or independently”; FR08
EquipmentIssue	A record that an equipment item was given out to a responsible person for an intended use, and of its condition when it came back.	EquipmentItem	“Before issue, staff confirm…”; “may return damaged…”; FR11, FR13
Inspection	A check of one equipment item, with its date, result and whether it is still open.	EquipmentItem	“awaiting inspection”; “unresolved inspections”; FR09
IncidentReport	A record of what staff saw in an incident or damage, without deciding the cause.	Vessel, EquipmentItem	“Staff need to record observations without deciding cause prematurely”; FR16

8.5 Trace retained concepts to evidence

Retained concept	Requirement, rule, scenario phrase or use case	Why the concept is needed
Berth	“Berths differ in length, depth, access, power and operational restrictions”; FR04, FR06; UC04	The harbour must know the details of each berth to choose a suitable one.
Vessel	“A request includes vessel, dimensions…”; FR01	Suitability depends on the size and needs of the vessel.
BerthRequest	FR01 to FR03, FR22; UC01 to UC03	Keeps the requested time, needs and provisional or confirmed status.
BerthChange	“Staff need to distinguish a changed berth from a rejected arrival”; FR07; UC05	Keeps what happened to a request, so changes agreed by radio reach the log.
HarbourNotice	“operational restrictions”; “works”; FR05; UC06	Shows when a berth cannot be used.
EquipmentCategory	“Different equipment categories require different authorisation or supervision”; FR12, BR04	Keeps the rules for each type of equipment.
EquipmentItem	“An item can be stored at the harbour but unavailable…”; FR09, FR10, BR07; UC08	Each item has its own condition, location and checks.
EquipmentReservation	“Equipment may be reserved with a berth or independently”; FR08, BR01; UC07	Keeps equipment for a vessel, with or without a berth.
EquipmentIssue	“Before issue, staff confirm the responsible person, intended use and current checks”; FR11, FR13, BR03; UC09, UC10	Shows who had the item and in what condition it came back.
Inspection	“awaiting inspection”; “unresolved inspections”; FR09; UC08	Shows if an item is checked, which decides if it is ready.

Retained concept	Requirement, rule, scenario phrase or use case	Why the concept is needed
IncidentReport	“Staff need to record observations without deciding cause prematurely”; FR16; UC13	Keeps what was seen so the cause can be found later.

8.6 Final candidate list

Retained	Berth, Vessel, BerthRequest, BerthChange, HarbourNotice, EquipmentCategory, EquipmentItem, EquipmentReservation, EquipmentIssue, Inspection, IncidentReport
Converted to attributes or values	length, depth, access, power, dimensions, vessel type, vessel condition, time window, purpose, contact, required services, status (provisional or confirmed), missing checks, arrival and departure times, access arrangement, completeness, compatibility, rated capacity, connector, condition, readiness, location, inspection status, responsible person, intended use, authorisation, supervision, observation, cause, weather, harbour conditions
Merged	request and booking -> BerthRequest; vessel details and vessel information -> Vessel; equipment and item -> EquipmentItem; temporary berth -> Berth; relocation, delay, delayed entry, cancellation, changed berth and rejected arrival -> BerthChange; harbour notice, restriction and works -> HarbourNotice; incident and damage -> IncidentReport
Removed as actors, operations, outputs or out of scope	vessel operator (for now), safety officer, harbour staff roles, overlap, berth conflict, delayed departure, early arrival, view, radio, telephone, logs, messages, navigation control, vessel maintenance, financial accounting, Kerry Harbour Services Cooperative, harbour, plan, berth use, issue, return, move, release, clear, check, block, reserve, change, allocate, confirm

9. Domain model

9.1 Selected attributes

Concept	Candidate attributes	Keep / remove	Reason
Berth	berthNumber, length, depth, access, power	Keep	Needed to choose a suitable berth (FR04). berthNumber tells berths apart, as the identifiers do in the lab examples.
Vessel	vesselName, vesselType, dimensions, condition	Keep	Suitability depends on the size of the vessel. Type and condition come from the scenario.
BerthRequest	requestDate, arrivalTime, departureTime, purpose, contact, requiredServices, status, missingChecks, actualArrival, actualDeparture, accessCleared	Keep	“A request includes…”; FR01-FR03, FR14, FR15, FR22.
BerthChange	changeType, reason, recordedAt, recordedBy	Keep	FR07. recordedAt and recordedBy come from NFR03 (who saved it and when).
HarbourNotice	noticeType, startDate, endDate	Keep	FR05: a restriction or works affects a berth for a period.
EquipmentCategory	categoryName, ratedCapacity, authorisationNeeded, supervisionNeeded	Keep	FR12. Rated capacity describes the type of equipment.
EquipmentItem	itemNumber, condition, readiness, location, complete	Keep	FR09, FR10, A02 (an incomplete set is unavailable).
EquipmentItem	ratedCapacity	Move	Moved to EquipmentCategory, because it describes the type, not one item.
EquipmentItem	connector, compatibility	Remove	No requirement needs them.
EquipmentItem	location	Keep as attribute	Berth areas have no details of their own, so location is a simple value (“Context matters”).
EquipmentReservation	startDate, endDate, status	Keep	FR08. A status as on Reservation in the lab.
EquipmentIssue	issueDate, returnDate, responsiblePerson, intendedUse, returnCondition, recordedAt, recordedBy	Keep	FR11, FR13, NFR03. Only the name of the responsible person and of the staff member who saved the record is needed, so they are simple values.
Inspection	inspectionDate, result, status	Keep	FR09; “unresolved inspections” (E22).
IncidentReport	observations, damage, cause, recordedAt, recordedBy	Keep	FR16, NFR03. The cause stays empty until it is decided (Q17).

9.2 Associations

Class A	Association name or sentence	Class B	Evidence	Keep?
BerthRequest	is for	Vessel	“A request includes vessel…”; FR01	Yes
BerthRequest	is allocated	Berth	FR06; UC04	Yes
BerthChange	changes	BerthRequest	FR07; UC05	Yes
BerthChange	moves to	Berth	“distinguish a changed berth from a rejected arrival”	Yes
HarbourNotice	affects	Berth	FR05; UC06	Yes
EquipmentCategory	describes	EquipmentItem	“Different equipment categories require…”; FR12	Yes
EquipmentReservation	reserves	EquipmentCategory	FR08; like Reservation requests Book in the lecture model	Yes
EquipmentReservation	is made with	BerthRequest	“Equipment may be reserved with a berth or independently”; FR08	Yes
EquipmentIssue	records issue of	EquipmentItem	FR13; like Loan records borrowing of BookCopy	Yes
Inspection	checks	EquipmentItem	“awaiting inspection”; FR09	Yes
IncidentReport	involves	Vessel	“Incidents… may involve vessel movement”; FR16	Yes
IncidentReport	involves	EquipmentItem	“…or equipment”; FR16	Yes
EquipmentIssue	fulfils	EquipmentReservation	The scenario does not say that an issue follows a reservation	No (speculative)
EquipmentReservation	is for	Vessel	The scenario does not say who an equipment reservation without a berth is for	No (speculative)

9.3 Multiplicities

Association	Multiplicity at Class A	Multiplicity at Class B	Reading from both ends	Evidence / rule
BerthRequest is for Vessel	0..*	1	Each berth request is for exactly one vessel. A vessel may have zero or many berth requests.	“A request includes vessel”; a vessel can come more than once
BerthRequest is allocated Berth	0..*	0..1	A berth request has zero or one berth (zero until a berth is allocated). A berth may be allocated to zero or many requests at different times.	FR03, FR06

Association	Multiplicity at Class A	Multiplicity at Class B	Reading from both ends	Evidence / rule
BerthChange changes BerthRequest	0..*	1	Each berth change belongs to exactly one request. A request may have zero or many changes.	FR07
BerthChange moves to Berth	0..*	0..1	A berth change moves the vessel to zero or one new berth (zero for a delay or refusal). A berth may receive zero or many changes.	“changed berth” vs “rejected arrival”
HarbourNotice affects Berth	0..*	1..*	A notice affects at least one berth. A berth may have zero or many notices.	FR05; no upper limit is given, so none is invented
EquipmentCategory describes EquipmentItem	1	1..*	Each item belongs to exactly one category. A category has at least one item.	“Equipment includes mobile fenders…”; like Book has copies 1..*
EquipmentReservation reserves EquipmentCategory	0..*	1	Each reservation is for exactly one category. A category may have zero or many reservations.	FR08; like Reservation 0..* to 1 Book
EquipmentReservation is made with BerthRequest	0..*	0..1	A reservation is made with zero or one berth request. A berth request may have zero or many reservations.	“with a berth or independently”
EquipmentIssue records issue of EquipmentItem	0..*	1	Each issue is of exactly one item. An item may have zero or many issues.	FR13; like Loan 0..* to 1 BookCopy
Inspection checks EquipmentItem	0..*	1	Each inspection checks exactly one item. An item may have zero or many inspections.	FR09
IncidentReport involves Vessel	0..*	0..*	A report may involve zero or many vessels. A vessel may be in zero or many reports.	“may involve”; no limit is given
IncidentReport involves EquipmentItem	0..*	0..*	A report may involve zero or many items. An item may be in zero or many reports.	“…or equipment”; no limit is given

9.4 Transactions and historical records

Information to retain	Best owning concept	Why
When the vessel wants to arrive and leave	BerthRequest	Part of the request (“arrival and departure window”).
If the request is provisional or confirmed, and which check is missing	BerthRequest	The status belongs to one request (FR03, FR22).
Which berth is allocated	Link between BerthRequest and Berth	A link between two concepts, not a value (FR06).

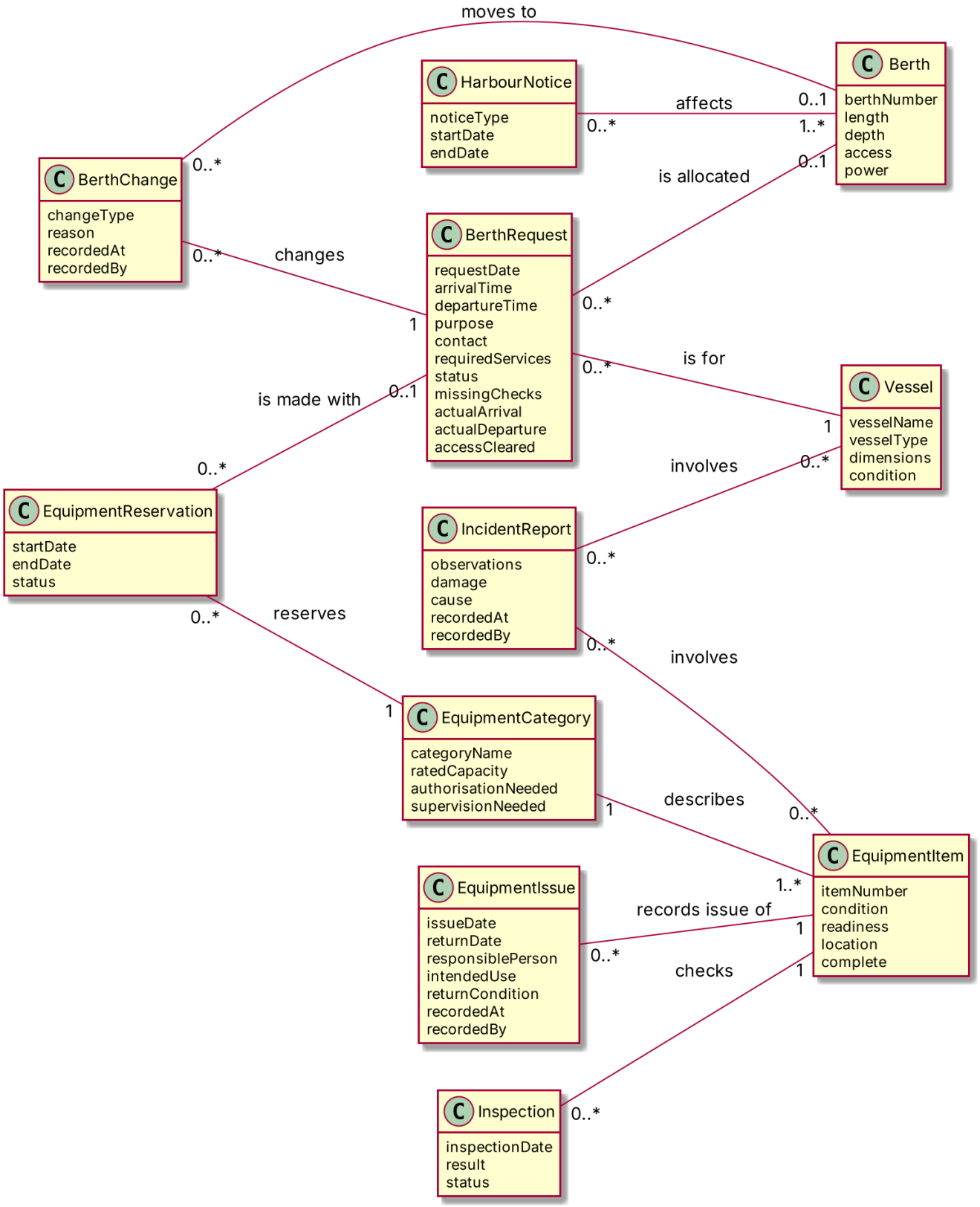
Information to retain	Best owning concept	Why
Move, delay or refusal, and who recorded it and when	BerthChange	One request can have several changes, each with its own type and date (FR07, NFR03).
When a berth cannot be used and why	HarbourNotice	A notice has its own period and affects berths (FR05).
Actual arrival, actual departure and access cleared	BerthRequest	They describe one visit (FR14, FR15).
If an item is ready to be given out	EquipmentItem	Condition, readiness and completeness belong to one item (FR09, FR10).
Which authorisation or supervision is needed	EquipmentCategory	It depends on the type of equipment (FR12).
For which period equipment is reserved	EquipmentReservation	A reservation has its own dates and status (FR08).
Who had the item, for what use, and its condition on return	EquipmentIssue	One issue keeps the history of one giving out (FR11, FR13).
When an item was checked, the result, and if it is still open	Inspection	Each check has its own date and result (FR09, E22).
What was seen in an incident, without the cause	IncidentReport	Observations are kept, the cause is decided later (FR16, Q17).

Decision: provisional hold

A provisional hold could be its own class. It is kept as a status of BerthRequest instead, because a hold has no other details in the scenario. The only extra detail would be how long the hold lasts, and that is not agreed (Q04). The request still keeps its request date, status and the missing check, so no information is lost.

9.5 Domain model diagram

Source file: diagrams/domain-model.puml.



9.6 Modelling rationale

Description and physical item. EquipmentCategory and EquipmentItem are kept apart, like Book and BookCopy in the lecture model. The category keeps the authorisation rules; the item keeps its own condition, location and checks.

Reservation and issue. EquipmentReservation is linked to a category and EquipmentIssue to one item, like Reservation requests Book and Loan records borrowing of BookCopy.

Transactions. BerthRequest, BerthChange, EquipmentReservation, EquipmentIssue and Inspection keep dates, status and history, because “transactions often deserve a class”.

Provisional hold as a status. A hold is a status of BerthRequest, not a class. The only extra data would be how long it lasts, and that is not agreed (Q04).

What is not in the model. Vessel operators and the safety officer are offstage (Section 3, A07). Conflicts, delayed departures and the coordinator’s views are worked out from other records. Staff are not classes: who saved a record is a simple value (recordedBy), like responsiblePerson. No methods, visibility, navigability, aggregation or foreign keys are used.

Multiplicity. Zero is used where participation is optional. No upper limits are invented: where the scenario gives no limit, * is used.

9.7 Review checklist

I checked the model with the review questions from the Session 6 lecture. Every class represents a meaningful domain concept: all 11 classes come from the glossary (Section 8.4), so each of them can be defined there. Important transactions and historical records are represented by BerthRequest, BerthChange, EquipmentReservation, EquipmentIssue and Inspection. The attributes are simple descriptive values (9.1).

Every association is supported by a requirement or use case (9.2), and two speculative links were removed. Every association shows multiplicity at both ends (9.3), and the association names can be read as domain sentences, for example “BerthRequest is for Vessel”. Software-design and database elements were removed: there are no methods, visibility, navigability, controllers or foreign keys. The model is readable at normal page size (9.5).

10. Traceability matrix

10.1 Traceability of Must Have requirements

Based on: Session 4 Lab, Task 9 (Business problem or stakeholder goal, Requirement, User story, Acceptance evidence, Use case). The CA1 brief also asks for links to Gherkin and domain concepts, so two columns are added.

Business problem or stakeholder goal	Requirement	User story	Acceptance evidence	Use case	Gherkin	Domain concept
P1: records are kept in separate logs and messages, so requests get lost	FR01	US01	A saved request with the vessel details, time, purpose, services and contact can be found and opened again	UC01 Record Berth Request	None	BerthRequest, Vessel
P5: plans change because of weather, vessel changes or restrictions	FR02	None	The new arrival time is shown on the request	UC02 Update Berth Request	None	BerthRequest
P3: there is no agreed rule for provisional holds	FR03	US02	The missing check is shown and the berth shows as held, not free and not confirmed	UC03 Confirm Berth Request	None	BerthRequest
P3: there is no agreed rule for provisional holds	FR22	None	A request with no missing check is confirmed; a request with a missing check stays provisional	UC03 Confirm Berth Request	None	BerthRequest
P2: an empty berth may be unsuitable or already reserved	FR04	US03	Only berths that fit the vessel and are free for the whole requested time are shown as suitable	UC04 Allocate Berth	GH01, GH02, GH03	Berth, Vessel, BerthRequest
P2: an empty berth may be unsuitable or already reserved	FR06	None	The chosen berth appears on the request and is shown as allocated for that time	UC04 Allocate Berth	GH01	BerthRequest, Berth
P5: it must be clear whether a vessel was moved, delayed or refused	FR07	US04	A move, a delay and a refusal are saved and shown as different outcomes	UC05 Record Berth Change	None	BerthChange
Harbour coordinator goal: reserve equipment for a vessel, with or without a berth (Section 3.2)	FR08	None	One booking is made with a berth and one without	UC07 Reserve Equipment	None	EquipmentReservation, EquipmentCategory, BerthRequest

Business problem or stakeholder goal	Requirement	User story	Acceptance evidence	Use case	Gherkin	Domain concept
P4 and P6: stored does not mean ready, and moves are not always recorded	FR09	None	The new status, checks and location of an item are saved	UC08 Update Equipment Status	None	EquipmentItem, Inspection
P4: stored does not mean complete, inspected and available	FR10	US05	Incomplete or unchecked equipment is shown as unavailable, with the reason	UC08 Update Equipment Status	None	EquipmentItem, Inspection
P4: incomplete, uninspected or unsuitable equipment may be issued	FR11	US06	Equipment cannot be given out until the responsible person, intended use and checks are confirmed	UC09 Issue Equipment	None	EquipmentIssue, EquipmentItem
P7: equipment may return damaged or incomplete	FR13	None	The record shows who had the item and its condition on return	UC09 Issue Equipment, UC10 Return Equipment	None	EquipmentIssue
Harbour assistant goal: save arrivals and departures (Section 3.2)	FR14	None	An arrival and a departure are recorded on the request	UC11 Record Vessel Arrival, UC12 Record Vessel Departure	None	BerthRequest
P7: it is unclear when a berth becomes available again	FR15	US07	The berth stays occupied until equipment and access are cleared	UC12 Record Vessel Departure	None	BerthRequest, Berth
P9: the coordinator needs one overview, but the information is in separate logs	FR17	US08	The overview shows provisional arrivals, conflicts, equipment readiness and location, restrictions, delayed departures and unresolved inspections	UC14 View Harbour Overview	None	Worked out from BerthRequest, Berth, EquipmentItem, HarbourNotice and Inspection (no class of its own, Section 8.3)
Data-protection officer: personal data must be handled properly (Section 3.1)	NFR01	US09	Only staff who are allowed to can see operator contact details	UC15 Grant Staff Access	None	BerthRequest (contact)
P1: records are easy to lose	NFR02	None	Nothing saved is lost after the system is restarted once	UC01, UC03 and UC08 (use cases that save these records)	None	BerthRequest, EquipmentItem
P6: changes agreed by radio may not reach the log	NFR03	None	Each berth change, equipment issue or return and incident note shows who saved it and when	UC05, UC09, UC10, UC13	None	BerthChange, EquipmentIssue, IncidentReport (recordedAt, recordedBy)

10.2 Coverage check

Every Must Have requirement (FR01 to FR04, FR06 to FR11, FR13 to FR15, FR17, FR22 and NFR01 to NFR03) is linked to a problem or stakeholder goal, at least one use case and at least one domain concept, so no Must Have requirement is isolated.

User stories were written only for selected requirements (Appendix A), so some rows have no story. These requirements are still covered by a use case and a way to check them (Section 4). Gherkin examples were written only for UC04, the detailed use case (Sections 6.3 and 7).

The Should and Could requirements (FR05, FR12, FR16) and NFR04 to NFR06 are traced in Sections 4 and 6. FR05 is also checked by GH04. FR17 has no class of its own, because the overview is worked out from other records (Section 8.3).

11. Reflection and AI declaration

11.1 Important decisions

Vessel operator as an offstage stakeholder. I decided that the vessel operator is an offstage stakeholder, not an actor, because harbour staff record the requests and changes for operators. The scenario does not say that operators use the system themselves, so this is assumption A07 and open question Q23. If Q23 is answered yes, I would add use cases for operators, such as sending a berth request, give them accounts and access rules (NFR01, UC15), and turn the operator's contact into a VesselOperator class (Section 8.3).

Provisional hold as a status. I kept the provisional hold as a status of BerthRequest instead of a separate class. In the scenario, a hold has no data of its own except how long it lasts, and staff have not agreed on that (E07, Q04). No information is lost, because the request still keeps its request date, its status and the missing check.

11.2 Revisions

- FR22 was added later. When I wrote UC03 Confirm Berth Request, I saw that a request has two outcomes, confirmed or still provisional, so confirming needed its own requirement.
- Vessel operator and safety officer were first listed as direct users in Section 3.1. They do not use the system, so I moved them to indirect stakeholders. I also changed the coordinator's expected interaction to match the use cases.
- Q27 was made narrower, because part of it repeated Q24, Q20 and Q05. I removed assumption A08 because it was not needed.
- recordedBy was first removed, because a person stored as a value looked like a foreign key. The traceability check showed that NFR03 needs who saved a record and when, so I added recordedBy back as a simple value.
- ratedCapacity was moved from EquipmentItem to EquipmentCategory, because it describes the type of equipment, not one item. Connector and compatibility were removed, because no requirement needs them.
- Two associations were rejected as speculative: EquipmentIssue fulfils EquipmentReservation, and EquipmentReservation is for Vessel. The scenario does not support them.

11.3 Limitations

The brief asks for about 8 to 12 functional requirements, but I have 18, because I kept each requirement atomic so it can be tested on its own. 15 of the 18 are Must, and these priorities should be checked with the coordinator and management. I wrote Gherkin examples only for UC04, so the other use cases are checked only through acceptance criteria and the “How to check” column. The numbers in NFR04 to NFR06 are proposals, because the scenario gives none.

11.4 Unresolved questions

The most important open question is Q04. How long a provisional hold should last? It decides when a berth becomes free again and what staff can tell the operator. The second is Q23, because if operators use the system themselves, the actors, the use cases and the domain model all change.

11.5 What I learned

I learned that the scenario, my assumptions and my questions must be kept apart, otherwise a guess starts to look like a fact. I also learned how useful traceability is. When I checked the matrix, I found that NFR03 needs to know who saved a record, but my model had no place for it, so I added recordedBy back. Next time I would check traceability earlier, not at the end.

11.6 AI declaration

I used Claude (Anthropic), during this CA1. I first read the scenario and wrote my own notes on the facts, problems, stakeholders and scope. I then used AI to translate my text into simple English, to explain lecture and lab material, and to check that the sections are consistent with each other. AI also suggested stakeholder questions and pointed out unclear parts of the scenario, which I compared with the scenario before keeping them. I discussed the domain model with AI, it explained options for classes, attributes, associations and multiplicities, and I made the final decisions. We discussed which use case to write in detail, and I chose UC04 Allocate Berth because it has the most rules. AI also checked that the sections are consistent and helped find gaps, for example in the traceability matrix.

Appendix A. User stories and acceptance criteria

A.1 Summary

ID	Title	Requirement	Use case	MoSCoW
US01	Save berth request	FR01	UC01	Must
US02	Mark request as provisional	FR03	UC03	Must
US03	Find a suitable berth	FR04	UC04	Must
US04	Save a berth change	FR07	UC05	Must
US05	Block unready equipment	FR10	UC08	Must
US06	Check before giving out equipment	FR11	UC09	Must
US07	Free berth only when cleared	FR15	UC12	Must
US08	See harbour overview	FR17	UC14	Must
US09	Protect operator contact details	NFR01	UC15	Must

A.2 Stories

US01 - Save berth request

Story	As a harbour assistant, I want to save a berth request with the vessel details, arrival and departure time and services needed, so that every request is written down in one place.
Business value / stakeholder goal	Requests are now spread across different logs and messages. One record stops requests getting lost.
Stakeholder	Harbour assistant
Related IDs	FR01, UC01
MoSCoW	Must
Acceptance criteria	1. A request can be saved with the vessel details, arrival and departure time, reason for the visit, services needed and operator contact. 2. The saved request can be found and opened again.
Questions revealed	Which vessel information and evidence is required before a booking is confirmed? Assumption: harbour staff enter requests for vessel operators.

US02 - Mark request as provisional

Story	As a harbour coordinator, I want to mark a request as provisional and see which check is still missing, so that I know which arrivals are not confirmed yet.
Business value / stakeholder goal	The coordinator can see which arrivals are not safe to rely on yet.
Stakeholder	Harbour coordinator
Related IDs	FR03, BR02, UC03
MoSCoW	Must
Acceptance criteria	1. A request can be marked as provisional when vessel information, insurance evidence or harbour approval is still missing. 2. The request shows which of these checks is still missing. 3. The berth for a provisional request is shown as held, not as free and not as confirmed. 4. A provisional request looks different from a confirmed request. 5. The provisional request appears in the coordinator's list of provisional arrivals.
Important limitation	The scenario does not say how long a provisional hold lasts. Therefore, the criteria should not say that the hold runs out after a set time.
Questions revealed	Which vessel information and evidence is required before a booking is confirmed? Is there a maximum duration for a provisional hold? What happens if evidence is not received? Who is allowed to mark a request as provisional or confirmed?

US03 - Find a suitable berth

Story	As a harbour coordinator, I want to see which berths fit the vessel and are free at that time, so that I don't give a vessel an empty berth that doesn't suit it.
Business value / stakeholder goal	Harbour coordinator: “An empty berth is not automatically suitable for the next vessel.”
Stakeholder	Harbour coordinator
Related IDs	FR04, UC04
MoSCoW	Must
Acceptance criteria	1. Only berths that are free for the whole requested time are shown. 2. A berth that is already booked or held for any part of that time is not shown as free. 3. A berth that is too short or not deep enough for the vessel is not shown as suitable. 4. A berth without a service the vessel needs (for example shore power) is not shown as suitable. 5. A berth affected by a restriction or works at that time is not shown as suitable. 6. For each berth shown, staff can see the berth details to check the result.

Given-When-Then example	Given a berth request for a vessel that needs shore power And the request has an arrival and departure time And berth A is free for that time, big enough and has shore power And berth B is free for that time but too short for the vessel When the coordinator looks for suitable berths Then berth A is shown as suitable And berth B is not shown as suitable
Important limitation	The scenario does not say exactly how suitability is decided. Therefore, the criteria only use the factors the scenario names: size, depth, services, bookings and restrictions.
Gherkin	GH01, GH02, GH03, GH04 (features/allocate-berth.feature)
Questions revealed	How should berth suitability for a vessel be determined? How is temporary allocation for an early arrival determined? Assumption: a provisional hold blocks the berth while checks are in progress.

US04 - Save a berth change

Story	As a harbour assistant, I want to save a berth change as a move, a delay or a refusal, so that changes agreed by radio also end up in the written log.
Business value / stakeholder goal	Harbour assistant: “A berth change agreed by radio may not reach the written log immediately.”
Stakeholder	Harbour assistant
Related IDs	FR07, UC05
MoSCoW	Must
Acceptance criteria	1. A change can be saved as a move, a delay or a refusal, and the three are shown as different outcomes. 2. The change is shown on the request it belongs to.
Questions revealed	Which conditions lead to relocation, delay or refusal? Who may relocate, delay or refuse an arrival?

US05 - Block unready equipment

Story	As an equipment officer, I want equipment that is incomplete or not checked to show as unavailable, so that nobody takes equipment that isn't ready.
Business value / stakeholder goal	Equipment officer: “Stored in the harbour does not mean complete, inspected and available.”

Stakeholder	Equipment officer
Related IDs	FR10, BR07, UC08
MoSCoW	Must
Acceptance criteria	1. Equipment marked as incomplete is shown as unavailable. 2. Equipment without a current check is shown as unavailable. 3. Unavailable equipment shows the reason it is unavailable.
Questions revealed	Assumption: an incomplete set is treated as unavailable.

US06 - Check before giving out equipment

Story	As an equipment officer, I want to confirm who is responsible, what the equipment is for and that the checks are up to date before I give it out, so that equipment only goes out for an agreed use.
Business value / stakeholder goal	Safety officer: “Equipment rating and inspection matter as much as physical availability.”
Stakeholder	Equipment officer (safety officer as offstage stakeholder)
Related IDs	FR11, BR03, UC09
MoSCoW	Must
Acceptance criteria	1. Equipment can't be recorded as given out until a responsible person is entered. 2. Equipment can't be recorded as given out until the intended use is entered. 3. Equipment can't be recorded as given out until staff confirm the checks are up to date. 4. Equipment shown as unavailable (incomplete or not checked) can't be given out. 5. If something is missing, staff can see what is missing. 6. When everything is confirmed, the issue is saved with the responsible person, the intended use and the time.
Given-When-Then example	Given an equipment item is complete and not shown as unavailable And a responsible person and the intended use are entered And staff confirm the checks are up to date When the equipment officer gives out the equipment Then the issue is saved with the responsible person, the intended use and the time
Questions revealed	Which equipment categories require which authorisation or supervision, and who grants it? Which checks are required, and how often must they be repeated to count as current? Who confirms that equipment is compatible with the vessel? What happens if issued equipment is unsuitable?

US07 - Free berth only when cleared

Story	As a harbour coordinator, I want a berth to become free only after the equipment and access are cleared, so that the next vessel doesn't get a berth that is still in use.
Business value / stakeholder goal	A berth that looks empty can still have equipment or access arrangements in place.
Stakeholder	Harbour coordinator
Related IDs	FR15, BR06, UC12
MoSCoW	Must
Acceptance criteria	1. After a departure is saved, the berth stays occupied while equipment or access is not cleared. 2. The berth is shown as free only after both are cleared.
Questions revealed	When do the berth and equipment become available again after departure?

US08 - See harbour overview

Story	As a harbour coordinator, I want to see provisional arrivals, berth conflicts, equipment readiness, restrictions, delayed departures and unresolved inspections in one place, so that I can plan without checking different logs.
Business value / stakeholder goal	The coordinator needs these views, but the information is now kept in separate logs and messages.
Stakeholder	Harbour coordinator
Related IDs	FR17, UC14
MoSCoW	Must
Acceptance criteria	1. The overview shows all provisional arrivals. 2. The overview shows berths with conflicting bookings or holds. 3. The overview shows the readiness and location of equipment. 4. The overview shows restrictions and works that affect berths. 5. The overview shows delayed departures. 6. The overview shows equipment with unresolved inspections.
Questions revealed	What counts as a berth conflict: only two bookings at the same time, or also a vessel placed in a berth that doesn't suit it? In what form does the coordinator need this information?

US09 - Protect operator contact details

Story	As a vessel operator, I want my contact details to be seen only by harbour staff who are allowed to, so that my personal information is protected.
Business value / stakeholder goal	Personal data must be handled properly (data-protection officer, Section 3).
Stakeholder	Vessel operator (offstage stakeholder, A07); data-protection officer
Related IDs	NFR01, UC15
MoSCoW	Must
Acceptance criteria	1. Staff who are allowed to can see operator contact details. 2. Staff who are not allowed to can't see operator contact details.
Questions revealed	Which staff roles are allowed to see operator contact details?
Note	The vessel operator is not an actor, but the story shows whose value is protected. The part of NFR01 about who may change records is traced through NFR01 directly.

A.3 Questions revealed

The acceptance criteria revealed these questions, so I recorded them instead of inventing rules. These questions should be clarified with stakeholders rather than answered through unsupported assumptions.

Most of the questions are already in Section 1.1 and Section 5. Three new questions came from writing the criteria and were added to Section 5 as Q24 (US02), Q25 (US08) and Q26 (US09).